

DANIYAL AHMAD KHAN

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Founder & Lead Engineer, Athena AI Solutions

PROFESSIONAL SUMMARY

Computer Vision and AI Engineer with 4+ years of experience delivering production-grade AI systems across sports analytics, healthcare imaging, edge AI, multi-camera vision, and AI-powered platforms. Experienced in end-to-end ownership from data pipelines and model training to real-time deployment on edge hardware and scalable cloud backends.

CORE SKILLS

- Computer Vision: Object Detection, Pose Estimation, Tracking, Action Recognition, Video Analytics
- Sports Biomechanics AI: Basketball and Golf
- Temporal Modeling: LSTM, sequence-based event detection
- Multi-Camera Systems: Calibration, Homography, Stitching, 3D/Top-Down analytics
- Edge AI: NVIDIA Jetson, DeepStream, TensorRT, CUDA, GStreamer, ONNX
- Backend & Cloud: FastAPI, Flask, AWS (Lambda, EC2, S3, API Gateway)
- LLMs & Agents: Tool-using agents, structured outputs
- Databases: PostgreSQL, DynamoDB, MongoDB

PROFESSIONAL EXPERIENCE

Athena AI Solutions — Founder & Lead AI Engineer

Sep 2021 – Present | Lahore, Pakistan

- Delivered production-grade computer vision, edge AI, and AI platform systems for startups and enterprises.
- Architected real-time video analytics pipelines optimized for latency and reliability on GPU hardware.
- Designed scalable backend APIs and cloud deployments supporting AI-driven dashboards and reporting.
- Led full project lifecycle from research and prototyping to deployment and optimization.

SELECTED PROJECT EXPERIENCE

Basketball Stance & Shot Event Detection System

Built a production CV pipeline combining object detection, pose estimation, and LSTM-based temporal modeling to detect shot phases and boundaries from game footage. Exported models to ONNX and deployed C++ inference pipelines for performance-critical environments. Tech: YOLOv8, MediaPipe, PyTorch, LSTM, ONNX, C++, OpenCV

Multi-Camera Vehicle Surround View & Stitching

Developed 180°/270°/360° image stitching using camera calibration and homography. Integrated RTSP fisheye camera streams and built real-time visualization and safety analytics dashboards. Tech: OpenCV, Homography, RTSP, Python, PyQt

3D Room Tracking with Dual PTZ Cameras

Implemented a DeepStream-based real-time detection and tracking pipeline with ONVIF PTZ auto-tracking and 3D/top-down OpenGL visualization. Optimized inference using CUDA and GStreamer. Tech: NVIDIA DeepStream, CUDA, GStreamer, OpenGL, PyQt, ONVIF

Mirror Vision Golf — AI Swing Analysis Platform

Designed swing phase detection and key-event extraction models for golf biomechanics. Optimized models for on-device inference and contributed to backend services for multi-device video synchronization. Tech: CoreML, Vision API, YOLOv8, LSTM, Swift, AWS

TripCorner — AI-Powered Travel Booking Assistant

Built a multi-agent AI system for multi-city travel planning using tool-based LLM execution and travel APIs. Deployed scalable backend services on AWS. Tech: GPT-4, FastAPI, React, AWS

EDUCATION

BSc Computer Science

Lahore University of Management Sciences (LUMS) | 2018 – 2022

TECHNICAL STACK

Python, C++, JavaScript | PyTorch, TensorFlow, OpenCV | FastAPI, Flask | NVIDIA Jetson, DeepStream, TensorRT | AWS | PostgreSQL, DynamoDB, MongoDB